COMP3710 | Project Feasibility Review

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1. User Need, Scope & Acceptance Criteria

This prototype aims to generate synthetic limit order book (LOB) sequences for quantitative researchers evaluating market simulations and risk strategies. The central dilemma is preserving market rules while reproducing real trading dynamics.

(1) Positive bid–ask spreads, ordered ten-level price ladders and non-negative volumes; (2) KL divergence ≤ 0.1 for spreads and mid-price returns; (3) depth-heatmap SSIM > 0.6; (4) baseline comparison, 3–5 failure-case audits and resource profiling.

2. Model Choice & Course Concepts

Data: LOBSTER AMZN Level-10 sample, 21 June 2012 (40 bid/ask price and volume features per observation). Model: TimeGAN with embedder, recovery, generator, supervisor and discriminator networks, combining adversarial learning with supervised temporal dynamics. Concepts: RNNs, GAN optimisation and distributional evaluation.

Planned pipeline: price log-returns, log-volume scaling, training-only normalisation, fixed-length sequences and chronological train/validation/test splits. An empirical resampling generator is the initial baseline.

3. Preliminary Feasibility Evidence — Rangpur

Data audit: 269,748 corresponding message and order-book records. No missing values, negative volumes, crossed/locked best quotes, price-ladder ordering violations or backward timestamps were found. The sample spans approximately 09:30–16:00.

Preprocessing / baseline: chronological 70/15/15 splitting produced 7,867 / 1,685 / 1,685 non-overlapping sequences (24 steps; 3 initial features), with no training NaNs. Spread resampling generated 40,463 positive spreads; mean generated spread was 0.14036 versus 0.09791 on held-out test data.

GPU smoke test: a simplified five-network prototype trained for a few steps on an NVIDIA A100 40GB with 512 training sequences; output shape (32, 24, 3), runtime 3.96 s and peak allocated GPU memory 27.15 MB. Outputs were finite. This verifies execution, not convergence or full order-book realism.

4. Risks, Compute Budget & Fallback

Risks include unstable GAN training, violated market constraints, limited single-day data coverage and higher full-model compute cost. Next, build full Level-10 sequences and TimeGAN training, compare against a full-book baseline, measure actual training resources and analyse 3–5 failures. Begin with short A100 allocations and small batches; reduce model size or sequence length if needed. Ask the tutor whether the free single-day sample is sufficient.